# Daniel Foreman-Mackey

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Member of Technical Staff, Anthropic, New York, NY

## Education

PhD 2015, Department of Physics, New York University. Advisor: Hogg
MSc 2010, Department of Physics, Queen's University, Canada. Advisor: Widrow
BSc 2008, Department of Physics, McGill University, Canada.

## Positions

Member of Technical Staff, Anthropic, 2025–present.
Research Engineer, Google DeepMind, 2024–2025.
Research Scientist, Flatiron Institute, 2022–2024.
Associate Research Scientist, Flatiron Institute, 2017–2022.
Sagan Postdoctoral Fellow, University of Washington, 2015–2017.

## Popular open-source software

**jax** — 36583 stars / 3989 forks
Composable transformations of Python+NumPy programs: differentiate, vectorize, JIT to GPU/TPU, and more [docs]

**tinygp** — 350 stars / 38 forks
The tiniest of Gaussian Process libraries [docs]

**emcee** — 1605 stars / 447 forks
The Python ensemble sampling toolkit for affine-invariant MCMC [docs]

**corner.py** — 578 stars / 238 forks
Make some beautiful corner plots [docs]

**exoplanet** — 242 stars / 57 forks
Fast & scalable MCMC for all your exoplanet needs! [docs]

**daft** — 686 stars / 123 forks
Render probabilistic graphical models using matplotlib [docs]

## Publications

refereed: 112 / first author: 9 / citations: 34,990 / h-index: 57 (2026-11-07)

### Refereed publications

112 Rubenzahl, Ryan A.; Hattori, Soichiro; Särkkä, Simo; Farr, Will M.; *et al.* (incl. **DFM**), 2026, *Scalable Gaussian Processes for Integrated and Overlapping Measurements via Augmented State-space Models*, The Astronomical Journal, **171**, 259 (arXiv:2601.02527) [2 citations]

111 Hapitas, Timothy; Widrow, Lawrence M.; Dharmawardena, Thavisha E.; & **Foreman-Mackey, Daniel**, 2025, *Gaussian Process Methods for Very Large Astrometric Data Sets*, The Astrophysical Journal, **995**, 35 (arXiv:2507.10317)

110 Chance, Quadry; **Foreman-Mackey, Daniel**; Ballard, Sarah; Casey, Andrew R.; *et al.*, 2025, *paired: A Statistical Framework for Detecting Stellar Binarity with Gaia RVs. I. Sensitivity to*

*Unresolved Binaries*, The Astrophysical Journal, **992**, 131 (arXiv:2206.11275) [15 citations]

109 Hattori, Soichiro; Angus, Ruth; **Foreman-Mackey, Daniel**; Lu, Yuxi (Lucy); & Colman, Isabel, 2025, *Measuring Long Stellar Rotation Periods (>10 days) from TESS FFI Light Curves is Possible: An Investigation Using TESS and ZTF*, The Astronomical Journal, **170**, 15 (arXiv:2505.10376) [10 citations]

108 Evans-Soma, Thomas M.; Sing, David K.; Barstow, Joanna K.; Piette, Anjali A. A.; *et al.* (incl. **DFM**), 2025, *SiO and a super-stellar C/O ratio in the atmosphere of the giant exoplanet WASP-121 b*, Nature Astronomy, **9**, 845 (arXiv:2506.01771) [33 citations]

107 Sayeed, Maryum; Angus, Ruth; Berger, Travis A.; Lu, Yuxi(Lucy); *et al.* (incl. **DFM**), 2025, *Exoplanet Occurrence Rate with Age for FGK Stars in Kepler*, The Astronomical Journal, **169**, 112 (arXiv:2501.13809) [9 citations]

106 Garrison, Lehman H.; **Foreman-Mackey, Daniel**; Shih, Yu-hsuan; & Barnett, Alex, 2024, *NIFTY-LS: Fast and Accurate Lomb–Scargle Periodograms Using a Non-uniform FFT*, Research Notes of the American Astronomical Society, **8**, 250 (arXiv:2409.08090) [16 citations]

105 Edwards, Thomas D. P.; Wong, Kaze W. K.; Lam, Kelvin K. H.; Coogan, Adam; *et al.* (incl. **DFM**), 2024, *Differentiable and hardware-accelerated waveforms for gravitational wave data analysis*, Physical Review D, **110**, 64028 (arXiv:2302.05329) [58 citations]

104 Dharmawardena, T. E.; Bailer-Jones, C. A. L.; Fouesneau, M.; **Foreman-Mackey, Daniel**; *et al.*, 2024, *All-sky three-dimensional dust density and extinction Maps of the Milky Way out to 2.8 kpc*, Monthly Notices of the Royal Astronomical Society, **532**, 3480 (arXiv:2406.06740) [29 citations]

103 Garcia, Lionel J.; **Foreman-Mackey, Daniel**; Murray, Catriona A.; Aigrain, Suzanne; *et al.*, 2024, *nuance: Efficient Detection of Planets Transiting Active Stars*, The Astronomical Journal, **167**, 284 (arXiv:2402.06835) [12 citations]

102 Fortune, Mark; Gibson, Neale P.; **Foreman-Mackey, Daniel**; Evans-Soma, Thomas M.; *et al.*, 2024, *How do wavelength correlations affect transmission spectra? Application of a new fast and flexible 2D Gaussian process framework to transiting exoplanet spectroscopy*, Astronomy and Astrophysics, **686** (arXiv:2402.15204) [10 citations]

101 Lu, Yuxi(Lucy); Angus, Ruth; **Foreman-Mackey, Daniel**; & Hattori, Soichiro, 2024, *In This Day and Age: An Empirical Gyrochronology Relation for Partially and Fully Convective Single Field Stars*, The Astronomical Journal, **167**, 159 (arXiv:2310.14990) [46 citations]

100 Yahalomi, Daniel A.; Angus, Ruth; Spergel, David N.; & **Foreman-Mackey, Daniel**, 2023, *Detecting Solar System Analogs through Joint Radial Velocity/Astrometric Surveys*, The Astronomical Journal, **166**, 258 (arXiv:2302.05064) [10 citations]

99 Dong, Jiayin; & **Foreman-Mackey, Daniel**, 2023, *A Hierarchical Bayesian Framework for Inferring the Stellar Obliquity Distribution*, The Astronomical Journal, **166**, 112 (arXiv:2305.14220) [58 citations]

98 Gagliano, Alexander; Contardo, Gabriella; **Foreman-Mackey, Daniel**; Malz, Alex I.; & Aleo, Patrick D., 2023, *First Impressions: Early-time Classification of Supernovae Using Host-galaxy Information and Shallow Learning*, The Astrophysical Journal, **954**, 6 (arXiv:2305.08894) [35 citations]

97 Aigrain, Suzanne; & **Foreman-Mackey, Daniel**, 2023, *Gaussian Process Regression for Astronomical Time Series*, Annual Review of Astronomy and Astrophysics, **61**, 329

(arXiv:2209.08940) [183 citations]

96 Blunt, Sarah; Carvalho, Adolfo; David, Trevor J.; Beichman, Charles; *et al.* (incl. **DFM**), 2023, *Overfitting Affects the Reliability of Radial Velocity Mass Estimates of the V1298 Tau Planets*, The Astronomical Journal, **166**, 62 (arXiv:2306.08145) [61 citations]

95 Tran, Quang H.; Bedell, Megan; **Foreman-Mackey, Daniel**; & Luger, Rodrigo, 2023, *Joint Modeling of Radial Velocities and Photometry with a Gaussian Process Framework*, The Astrophysical Journal, **950**, 162 (arXiv:2305.00988) [16 citations]

94 Wong, Kaze W. K.; Gabrié, Marylou; & **Foreman-Mackey, Daniel**, 2023, *flowMC: Normalizing flow enhanced sampling package for probabilistic inference in JAX*, The Journal of Open Source Software, **8**, 5021 (arXiv:2211.06397) [56 citations]

93 Alderson, Lili; Wakeford, Hannah R.; Alam, Munazza K.; Batalha, Natasha E.; *et al.* (incl. **DFM**), 2023, *Early Release Science of the exoplanet WASP-39b with JWST NIRSpec G395H*, Nature, **614**, 664 (arXiv:2211.10488) [307 citations]

92 Mikal-Evans, Thomas; Sing, David K.; Dong, Jiayin; **Foreman-Mackey, Daniel**; *et al.*, 2023, *A JWST NIRSpec Phase Curve for WASP-121b: Dayside Emission Strongest Eastward of the Substellar Point and Nightside Conditions Conducive to Cloud Formation*, The Astrophysical Journal Letters, **943** (arXiv:2301.03209) [52 citations]

91 Dharmawardena, T. E.; Bailer-Jones, C. A. L.; Fouesneau, M.; **Foreman-Mackey, Daniel**; *et al.*, 2023, *The three-dimensional structure of galactic molecular cloud complexes out to 2.5 kpc*, Monthly Notices of the Royal Astronomical Society, **519**, 228 (arXiv:2210.03615) [20 citations]

90 Jo, Yongseok; Genel, Shy; Wandelt, Benjamin; Somerville, Rachel S.; *et al.* (incl. **DFM**), 2023, *Calibrating Cosmological Simulations with Implicit Likelihood Inference Using Galaxy Growth Observables*, The Astrophysical Journal, **944**, 67 (arXiv:2211.16461) [23 citations]

89 Brande, Jonathan; Crossfield, Ian J. M.; Kreidberg, Laura; Oklopčić, Antonija; *et al.* (incl. **DFM**), 2022, *A Mirage or an Oasis? Water Vapor in the Atmosphere of the Warm Neptune TOI-674 b*, The Astronomical Journal, **164**, 197 (arXiv:2201.04197) [24 citations]

88 Nagaraj, Gautam; Forbes, John C.; Leja, Joel; **Foreman-Mackey, Daniel**; & Hayward, Christopher C., 2022, *Empirical Dust Attenuation Model Leads to More Realistic UVJ Diagram for TNG100 Galaxies*, The Astrophysical Journal, **939**, 29 (arXiv:2204.06449) [7 citations]

87 Eilers, Anna-Christina; Hogg, David W.; Schölkopf, Bernhard; **Foreman-Mackey, Daniel**; *et al.*, 2022, *A Generative Model for Quasar Spectra*, The Astrophysical Journal, **938**, 17 (arXiv:2209.02725) [10 citations]

86 Farrell, Eoin; Jermyn, Adam S.; Cantiello, Matteo; & **Foreman-Mackey, Daniel**, 2022, *The Initial Magnetic Field Distribution in AB Stars*, The Astrophysical Journal, **938**, 10 (arXiv:2210.11180) [5 citations]

85 Astropy Collaboration; Price-Whelan, Adrian M.; Lim, Pey Lian; Earl, Nicholas; *et al.* (incl. **DFM**), 2022, *The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package*, The Astrophysical Journal, **935**, 167 (arXiv:2206.14220) [6353 citations]

84 Luger, Rodrigo; Agol, Eric; Bartolić, Fran; & **Foreman-Mackey, Daniel**, 2022, *Analytic Light Curves in Reflected Light: Phase Curves, Occultations, and Non-Lambertian Scattering for Spherical Planets and Moons*, The Astronomical Journal, **164**, 4 (arXiv:2103.06275) [17

citations]

83 Angus, Ruth; Price-Whelan, Adrian M.; Zinn, Joel C.; Bedell, Megan; *et al.* (incl. **DFM**), 2022, *The 3D Galactocentric Velocities of Kepler Stars: Marginalizing Over Missing Radial Velocities*, The Astronomical Journal, **164**, 25 (arXiv:2205.08901) [5 citations]

82 Hattori, Soichiro; **Foreman-Mackey, Daniel**; Hogg, David W.; Montet, Benjamin T.; *et al.*, 2022, *The unpopular Package: A Data-driven Approach to Detrending TESS Full-frame Image Light Curves*, The Astronomical Journal, **163**, 284 (arXiv:2106.15063) [69 citations]

81 Johnson, Marshall C.; David, Trevor J.; Petigura, Erik A.; Isaacson, Howard T.; *et al.* (incl. **DFM**), 2022, *An Aligned Orbit for the Young Planet V1298 Tau b*, The Astronomical Journal, **163**, 247 (arXiv:2110.10707) [41 citations]

80 Nagaraj, Gautam; Forbes, John C.; Leja, Joel; **Foreman-Mackey, Daniel**; & Hayward, Christopher C., 2022, *A Bayesian Population Model for the Observed Dust Attenuation in Galaxies*, The Astrophysical Journal, **932**, 54 (arXiv:2202.05102) [39 citations]

79 Powell, Brian P.; Kruse, Ethan; Montet, Benjamin T.; Feinstein, Adina D.; *et al.* (incl. **DFM**), 2022, *The NASA GSFC TESS Full Frame Image Light Curve Data Set*, Research Notes of the American Astronomical Society, **6**, 111 [31 citations]

78 Hitchcock, J. A.; Bramich, D. M.; **Foreman-Mackey, Daniel**; Hogg, David W.; & Hundertmark, M., 2022, *The Thresher: Lucky imaging without the waste*, Monthly Notices of the Royal Astronomical Society, **511**, 5372 (arXiv:2202.04686)

77 Bartolić, Fran; Luger, Rodrigo; **Foreman-Mackey, Daniel**; Howell, Robert R.; & Rathbun, Julie A., 2022, *Occultation Mapping of Io's Surface in the Near-infrared. I. Inferring Static Maps*, The Planetary Science Journal, **3**, 67 (arXiv:2103.03758) [5 citations]

76 Dharmawardena, T. E.; Bailer-Jones, C. A. L.; Fouesneau, M.; & **Foreman-Mackey, Daniel**, 2022, *Three-dimensional dust density structure of the Orion, Cygnus X, Taurus, and Perseus star-forming regions*, Astronomy and Astrophysics, **658** (arXiv:2111.06672) [27 citations]

75 Feinstein, Adina D.; David, Trevor J.; Montet, Benjamin T.; **Foreman-Mackey, Daniel**; *et al.*, 2022, *V1298 Tau with TESS: Updated Ephemerides, Radii, and Period Constraints from a Second Transit of V1298 Tau e*, The Astrophysical Journal Letters, **925** (arXiv:2111.08660) [29 citations]

74 Martin, David V.; El-Badry, Kareem; Hodžić, Vedad Kunovac; Triaud, Amaury H. M. J.; *et al.* (incl. **DFM**), 2021, *TOI-1259Ab - a gas giant planet with 2.7 per cent deep transits and a bound white dwarf companion*, Monthly Notices of the Royal Astronomical Society, **507**, 4132 (arXiv:2101.02707) [20 citations]

73 Van Eylen, V.; Astudillo-Defru, N.; Bonfils, X.; Livingston, J.; *et al.* (incl. **DFM**), 2021, *Masses and compositions of three small planets orbiting the nearby M dwarf L231-32 (TOI-270) and the M dwarf radius valley*, Monthly Notices of the Royal Astronomical Society, **507**, 2154 (arXiv:2101.01593) [149 citations]

72 Gan, Tianjun; Bedell, Megan; Wang, Sharon Xuesong; **Foreman-Mackey, Daniel**; *et al.*, 2021, *HD 183579b: a warm sub-Neptune transiting a solar twin detected by TESS*, Monthly Notices of the Royal Astronomical Society, **507**, 2220 (arXiv:2107.14015) [8 citations]

71 Luger, Rodrigo; **Foreman-Mackey, Daniel**; & Hedges, Christina, 2021, *Mapping Stellar Surfaces. II. An Interpretable Gaussian Process Model for Light Curves*, The Astronomical Journal, **162**, 124 (arXiv:2102.01697) [42 citations]

70 Luger, Rodrigo; **Foreman-Mackey, Daniel**; Hedges, Christina; & Hogg, David W., 2021,

*Mapping Stellar Surfaces. I. Degeneracies in the Rotational Light-curve Problem*, The Astronomical Journal, **162**, 123 (arXiv:2102.00007) [65 citations]

69 Dong, Jiayin; Huang, Chelsea X.; Dawson, Rebekah I.; **Foreman-Mackey, Daniel**; *et al.*, 2021, *Warm Jupiters in TESS Full-frame Images: A Catalog and Observed Eccentricity Distribution for Year 1*, The Astrophysical Journal Supplement Series, **255**, 6 (arXiv:2104.01970) [51 citations]

68 Luger, Rodrigo; **Foreman-Mackey, Daniel**; & Hedges, Christina, 2021, *starry_process: Interpretable Gaussian processes for stellar light curves*, The Journal of Open Source Software, **6**, 3071 (arXiv:2102.01774) [7 citations]

67 Hitchcock, James A.; Hundertmark, Markus; **Foreman-Mackey, Daniel**; Bachelet, Etienne; *et al.*, 2021, *PyTorchDIA: a flexible, GPU-accelerated numerical approach to Difference Image Analysis*, Monthly Notices of the Royal Astronomical Society, **504**, 3561 (arXiv:2104.13715) [7 citations]

66 **Foreman-Mackey, Daniel**; Luger, Rodrigo; Agol, Eric; Barclay, Thomas; *et al.*, 2021, *exoplanet: Gradient-based probabilistic inference for exoplanet data & other astronomical time series*, The Journal of Open Source Software, **6**, 3285 (arXiv:2105.01994) [246 citations]

65 David, Trevor J.; Contardo, Gabriella; Sandoval, Angeli; Angus, Ruth; *et al.* (incl. **DFM**), 2021, *Evolution of the Exoplanet Size Distribution: Forming Large Super-Earths Over Billions of Years*, The Astronomical Journal, **161**, 265 (arXiv:2011.09894) [63 citations]

64 Gordon, Tyler A.; Davenport, James R. A.; Angus, Ruth; **Foreman-Mackey, Daniel**; *et al.*, 2021, *Stellar Rotation in the K2 Sample: Evidence for Modified Spin-down*, The Astrophysical Journal, **913**, 70 (arXiv:2101.07886) [56 citations]

63 Tamayo, Daniel; Gilbertson, Christian; & **Foreman-Mackey, Daniel**, 2021, *Stability constrained characterization of multiplanet systems*, Monthly Notices of the Royal Astronomical Society, **501**, 4798 (arXiv:2009.11831) [16 citations]

62 Agol, Eric; Dorn, Caroline; Grimm, Simon L.; Turbet, Martin; *et al.* (incl. **DFM**), 2021, *Refining the Transit-timing and Photometric Analysis of TRAPPIST-1: Masses, Radii, Densities, Dynamics, and Ephemerides*, The Planetary Science Journal, **2**, 1 (arXiv:2010.01074) [378 citations]

61 Hedges, Christina; Luger, Rodrigo; Dotson, Jessie; **Foreman-Mackey, Daniel**; & Barentsen, Geert, 2021, *Multiwavelength Photometry Derived from Monochromatic Kepler Data*, The Astronomical Journal, **161**, 95 (arXiv:2102.00044) [7 citations]

60 Gordon, Tyler A.; Agol, Eric; & **Foreman-Mackey, Daniel**, 2020, *A Fast, Two-dimensional Gaussian Process Method Based on Celerite: Applications to Transiting Exoplanet Discovery and Characterization*, The Astronomical Journal, **160**, 240 (arXiv:2007.05799) [28 citations]

59 Villaume, Alexa; **Foreman-Mackey, Daniel**; Romanowsky, Aaron J.; Brodie, Jean; & Strader, Jay, 2020, *The Assembly History of M87 through Radial Variations in Chemical Abundances of Its Field Star and Globular Cluster Populations*, The Astrophysical Journal, **900**, 95 (arXiv:2006.16280) [10 citations]

58 Angus, Ruth; Beane, Angus; Price-Whelan, Adrian M.; Newton, Elisabeth; *et al.* (incl. **DFM**), 2020, *Exploring the Evolution of Stellar Rotation Using Galactic Kinematics*, The Astronomical Journal, **160**, 90 (arXiv:2005.09387) [54 citations]

57 Plavchan, Peter; Barclay, Thomas; Gagné, Jonathan; Gao, Peter; *et al.* (incl. **DFM**), 2020, *Publisher Correction: A planet within the debris disk around the pre-main-sequence star AU*

*Microscopii*, Nature, **583** [2 citations]

56 Hey, Daniel; Murphy, Simon; **Foreman-Mackey, Daniel**; Bedding, Timothy; *et al.*, 2020, *Maelstrom: A Python package for identifying companions to pulsating stars from their light travel time variations*, The Journal of Open Source Software, **5**, 2125 [8 citations]

55 Plavchan, Peter; Barclay, Thomas; Gagné, Jonathan; Gao, Peter; *et al.* (incl. **DFM**), 2020, *A planet within the debris disk around the pre-main-sequence star AU Microscopii*, Nature, **582**, 497 (arXiv:2006.13248) [246 citations]

54 Hey, Daniel R.; Murphy, Simon J.; **Foreman-Mackey, Daniel**; Bedding, Timothy R.; *et al.*, 2020, *Forward Modeling the Orbits of Companions to Pulsating Stars from Their Light Travel Time Variations*, The Astronomical Journal, **159**, 202 (arXiv:2003.02379) [23 citations]

53 Agol, Eric; Luger, Rodrigo; & **Foreman-Mackey, Daniel**, 2020, *Analytic Planetary Transit Light Curves and Derivatives for Stars with Polynomial Limb Darkening*, The Astronomical Journal, **159**, 123 (arXiv:1908.03222) [169 citations]

52 Gillen, Edward; Briegal, Joshua T.; Hodgkin, Simon T.; **Foreman-Mackey, Daniel**; *et al.*, 2020, *NGTS clusters survey - I. Rotation in the young benchmark open cluster Blanco 1*, Monthly Notices of the Royal Astronomical Society, **492**, 1008 (arXiv:1911.09705) [64 citations]

51 **Foreman-Mackey, Daniel**; Farr, Will; Sinha, Manodeep; Archibald, Anne; *et al.*, 2019, *emcee v3: A Python ensemble sampling toolkit for affine-invariant MCMC*, The Journal of Open Source Software, **4**, 1864 (arXiv:1911.07688) [307 citations]

50 David, Trevor J.; Petigura, Erik A.; Luger, Rodrigo; **Foreman-Mackey, Daniel**; *et al.*, 2019, *Four Newborn Planets Transiting the Young Solar Analog V1298 Tau*, The Astrophysical Journal Letters, **885** (arXiv:1910.04563) [168 citations]

49 Angus, Ruth; Morton, Timothy D.; **Foreman-Mackey, Daniel**; van Saders, Jennifer; *et al.*, 2019, *Toward Precise Stellar Ages: Combining Isochrone Fitting with Empirical Gyrochronology*, The Astronomical Journal, **158**, 173 (arXiv:1908.07528) [138 citations]

48 Bedell, Megan; Hogg, David W.; **Foreman-Mackey, Daniel**; Montet, Benjamin T.; & Luger, Rodrigo, 2019, *WOBBLE: A Data-driven Analysis Technique for Time-series Stellar Spectra*, The Astronomical Journal, **158**, 164 (arXiv:1901.00503) [60 citations]

47 Feinstein, Adina D.; Montet, Benjamin T.; **Foreman-Mackey, Daniel**; Bedell, Megan E.; *et al.*, 2019, *eleanor: An Open-source Tool for Extracting Light Curves from the TESS Full-frame Images*, Publications of the Astronomical Society of the Pacific, **131**, 94502 (arXiv:1903.09152) [265 citations]

46 Kruse, Ethan; Agol, Eric; Luger, Rodrigo; & **Foreman-Mackey, Daniel**, 2019, *Detection of Hundreds of New Planet Candidates and Eclipsing Binaries in K2 Campaigns 0-8*, The Astrophysical Journal Supplement Series, **244**, 11 (arXiv:1907.10806) [71 citations]

45 Angus, Ruth; Morton, Timothy; & **Foreman-Mackey, Daniel**, 2019, *stardate: Combining dating methods for better stellar ages*, The Journal of Open Source Software, **4**, 1469 [19 citations]

44 Kostov, Veselin B.; Schlieder, Joshua E.; Barclay, Thomas; Quintana, Elisa V.; *et al.* (incl. **DFM**), 2019, *The L 98-59 System: Three Transiting, Terrestrial-size Planets Orbiting a Nearby M Dwarf*, The Astronomical Journal, **158**, 32 (arXiv:1903.08017) [126 citations]

43 Siemiginowska, Aneta; Eadie, Gwendolyn; Czekala, Ian; Feigelson, Eric; *et al.* (incl. **DFM**), 2019, *The Next Decade of Astroinformatics and Astrostatistics*, Bulletin of the American

Astronomical Society, **51**, 355 (arXiv:1903.06796) [11 citations]

42 Luger, Rodrigo; Agol, Eric; **Foreman-Mackey, Daniel**; Fleming, David P.; *et al.*, 2019, *starry: Analytic Occultation Light Curves*, The Astronomical Journal, **157**, 64 (arXiv:1810.06559) [331 citations]

41 Van Eylen, Vincent; Albrecht, Simon; Huang, Xu; MacDonald, Mariah G.; *et al.* (incl. **DFM**), 2019, *The Orbital Eccentricity of Small Planet Systems*, The Astronomical Journal, **157**, 61 (arXiv:1807.00549) [257 citations]

40 Brewer, John M.; Wang, Songhu; Fischer, Debra A.; & **Foreman-Mackey, Daniel**, 2018, *Compact Multi-planet Systems are more Common around Metal-poor Hosts*, The Astrophysical Journal Letters, **867** (arXiv:1810.10009) [58 citations]

39 Ness, Melissa K.; Silva Aguirre, Victor; Lund, Mikkel N.; Cantiello, Matteo; *et al.* (incl. **DFM**), 2018, *Inference of Stellar Parameters from Brightness Variations*, The Astrophysical Journal, **866**, 15 (arXiv:1805.04519) [15 citations]

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37 Luger, Rodrigo; Kruse, Ethan; **Foreman-Mackey, Daniel**; Agol, Eric; & Saunders, Nicholas, 2018, *An Update to the EVEREST K2 Pipeline: Short Cadence, Saturated Stars, and Kepler-like Photometry Down to Kp = 15*, The Astronomical Journal, **156**, 99 (arXiv:1702.05488) [166 citations]

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34 Hogg, David W.; & **Foreman-Mackey, Daniel**, 2018, *Data Analysis Recipes: Using Markov Chain Monte Carlo*, The Astrophysical Journal Supplement Series, **236**, 11 (arXiv:1710.06068) [308 citations]

33 Angus, Ruth; Morton, Timothy; Aigrain, Suzanne; **Foreman-Mackey, Daniel**; & Rajpaul, Vinesh, 2018, *Inferring probabilistic stellar rotation periods using Gaussian processes*, Monthly Notices of the Royal Astronomical Society, **474**, 2094 (arXiv:1706.05459) [228 citations]

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30 Grunblatt, Samuel K.; Huber, Daniel; Gaidos, Eric; Lopez, Eric D.; *et al.* (incl. **DFM**), 2017, *Seeing Double with K2: Testing Re-inflation with Two Remarkably Similar Planets around Red Giant Branch Stars*, The Astronomical Journal, **154**, 254 (arXiv:1706.05865) [89 citations]

29 Montet, Benjamin T.; Tovar, Guadalupe; & **Foreman-Mackey, Daniel**, 2017, *Long-term Photometric Variability in Kepler Full-frame Images: Magnetic Cycles of Sun-like Stars*, The Astrophysical Journal, **851**, 116 (arXiv:1705.07928) [99 citations]

28 Luger, Rodrigo; **Foreman-Mackey, Daniel**; & Hogg, David W., 2017, *Linear Models for*

*Systematics and Nuisances*, Research Notes of the American Astronomical Society, **1**, 7 (arXiv:1710.11136) [18 citations]

27 Price-Whelan, Adrian M.; & **Foreman-Mackey, Daniel**, 2017, *schwimmbad: A uniform interface to parallel processing pools in Python*, The Journal of Open Source Software, **2**, 357 [52 citations]

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25 Price-Whelan, Adrian M.; Hogg, David W.; **Foreman-Mackey, Daniel**; & Rix, Hans-Walter, 2017, *The Joker: A Custom Monte Carlo Sampler for Binary-star and Exoplanet Radial Velocity Data*, The Astrophysical Journal, **837**, 20 (arXiv:1610.07602) [154 citations]

24 **Foreman-Mackey, Daniel**; Morton, Timothy D.; Hogg, David W.; Agol, Eric; & Schölkopf, Bernhard, 2016, *The Population of Long-period Transiting Exoplanets*, The Astronomical Journal, **152**, 206 (arXiv:1607.08237) [94 citations]

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21 Luger, Rodrigo; Agol, Eric; Kruse, Ethan; Barnes, Rory; *et al.* (incl. **DFM**), 2016, *EVEREST: Pixel Level Decorrelation of K2 Light Curves*, The Astronomical Journal, **152**, 100 (arXiv:1607.00524) [269 citations]

20 Angus, Ruth; Aigrain, Suzanne; & **Foreman-Mackey, Daniel**, 2016, *Stellar rotation period inference with Gaussian processes*, IAU Focus Meeting, **29A**, 191

19 Wang, Dun; Hogg, David W.; **Foreman-Mackey, Daniel**; & Schölkopf, Bernhard, 2016, *A Causal, Data-driven Approach to Modeling the Kepler Data*, Publications of the Astronomical Society of the Pacific, **128**, 94503 (arXiv:1508.01853) [38 citations]

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11 Angus, Ruth; Aigrain, Suzanne; **Foreman-Mackey, Daniel**; & McQuillan, Amy, 2015, *Calibrating gyrochronology using Kepler asteroseismic targets*, Monthly Notices of the Royal Astronomical Society, **450**, 1787 (arXiv:1502.06965) [199 citations]

10 **Foreman-Mackey, Daniel**; Montet, Benjamin T.; Hogg, David W.; Morton, Timothy D.; *et al.*, 2015, *A Systematic Search for Transiting Planets in the K2 Data*, The Astrophysical Journal, **806**, 215 (arXiv:1502.04715) [125 citations]

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7 Barclay, Thomas; Endl, Michael; Huber, Daniel; **Foreman-Mackey, Daniel**; *et al.*, 2015, *Radial Velocity Observations and Light Curve Noise Modeling Confirm that Kepler-91b is a Giant Planet Orbiting a Giant Star*, The Astrophysical Journal, **800**, 46 (arXiv:1408.3149) [77 citations]

6 **Foreman-Mackey, Daniel**; Hogg, David W.; & Morton, Timothy D., 2014, *Exoplanet Population Inference and the Abundance of Earth Analogs from Noisy, Incomplete Catalogs*, The Astrophysical Journal, **795**, 64 (arXiv:1406.3020) [278 citations]

5 Dawson, Rebekah I.; Johnson, John Asher; Fabrycky, Daniel C.; **Foreman-Mackey, Daniel**; *et al.*, 2014, *Large Eccentricity, Low Mutual Inclination: The Three-dimensional Architecture of a Hierarchical System of Giant Planets*, The Astrophysical Journal, **791**, 89 (arXiv:1405.5229) [87 citations]

4 Dorman, Claire E.; Widrow, Lawrence M.; Guhathakurta, Puragra; Seth, Anil C.; *et al.* (incl. **DFM**), 2013, *A New Approach to Detailed Structural Decomposition from the SPLASH and PHAT Surveys: Kicked-up Disk Stars in the Andromeda Galaxy?*, The Astrophysical Journal, **779**, 103 (arXiv:1310.4179) [57 citations]

3 Brewer, Brendon J.; **Foreman-Mackey, Daniel**; & Hogg, David W., 2013, *Probabilistic Catalogs for Crowded Stellar Fields*, The Astronomical Journal, **146**, 7 (arXiv:1211.5805) [48 citations]

2 **Foreman-Mackey, Daniel**; Hogg, David W.; Lang, Dustin; & Goodman, Jonathan, 2013, *emcee: The MCMC Hammer*, Publications of the Astronomical Society of the Pacific, **125**, 306 (arXiv:1202.3665) [13948 citations]

1 Weisz, Daniel R.; Fouesneau, Morgan; Hogg, David W.; Rix, Hans-Walter; *et al.* (incl. **DFM**), 2013, *The Panchromatic Hubble Andromeda Treasury. IV. A Probabilistic Approach to Inferring the High-mass Stellar Initial Mass Function and Other Power-law Functions*, The Astrophysical Journal, **762**, 123 (arXiv:1211.6105) [36 citations]

### Preprints & white papers

9 Hey, Daniel; Huber, Daniel; Ong, Joel; Stello, Dennis; & **Foreman-Mackey, Daniel**, 2024, *Precise Time-Domain Asteroseismology and a Revised Target List for TESS Solar-Like Oscillators*, ArXiv (arXiv:2403.02489) [18 citations]

8 Cabezas, Alberto; Corenflos, Adrien; Lao, Junpeng; Louf, Rémi; *et al.* (incl. **DFM**), 2024, *BlackJAX: Composable Bayesian inference in JAX*, ArXiv (arXiv:2402.10797) [115 citations]

7 Blanton, Michael R.; Evans, Janet D.; Norman, Dara; O'Mullane, William; *et al.* (incl. **DFM**), 2023, *The Future of Astronomical Data Infrastructure: Meeting Report*, ArXiv (arXiv:2311.04272) [2 citations]

6 Eadie, Gwendolyn M.; Speagle, Joshua S.; Cisewski-Kehe, Jessi; **Foreman-Mackey, Daniel**; *et al.*, 2023, *Practical Guidance for Bayesian Inference in Astronomy*, ArXiv (arXiv:2302.04703) [16 citations]

5 Luger, Rodrigo; Bedell, Megan; **Foreman-Mackey, Daniel**; Crossfield, Ian J. M.; *et al.*, 2021, *Mapping stellar surfaces III: An Efficient, Scalable, and Open-Source Doppler Imaging Model*, ArXiv (arXiv:2110.06271) [70 citations]

4 Wang, Dun; Hogg, David W.; **Foreman-Mackey, Daniel**; & Schölkopf, Bernhard, 2017, *A pixel-level model for event discovery in time-domain imaging*, ArXiv (arXiv:1710.02428) [13 citations]

3 Barnes, Rory; Deitrick, Russell; Luger, Rodrigo; Driscoll, Peter E.; *et al.* (incl. **DFM**), 2016, *The Habitability of Proxima Centauri b I: Evolutionary Scenarios*, ArXiv (arXiv:1608.06919) [70 citations]

2 Montet, Benjamin T.; Angus, Ruth; Barclay, Tom; Dawson, Rebekah; *et al.* (incl. **DFM**), 2013, *Maximizing Kepler science return per telemetered pixel: Searching the habitable zones of the brightest stars*, ArXiv (arXiv:1309.0654)

1 Hogg, David W.; Angus, Ruth; Barclay, Tom; Dawson, Rebekah; *et al.* (incl. **DFM**), 2013, *Maximizing Kepler science return per telemetered pixel: Detailed models of the focal plane in the two-wheel era*, ArXiv (arXiv:1309.0653)

## Mentorship

I collaborate with and mentor many students and postdocs, often on a single project. Below is a list of the group members who I have formally mentored as part of the Flatiron Research Fellowship and Pre-doctoral Fellowship at the Center for Computational Astrophysics.

*Current postdocs:* Thavisha Dharmawardena, Jiayin Dong, Nora Eisner, Lionel Garcia, Joseph Long.

*Current students:* Quadry Chance, Soichiro Hattori.

*Former postdocs:* Megan Bedell, Trevor David, Rodrigo Luger.

*Former students:* Fran Bartolić, Eoin Farrell, Alex Gagliano, Karl Jaehnig, Gautam Nagaraj, Pa Chia Thao, Nhat Quang Hoang Tran.

## Selected invited talks & tutorials

*Open software for Astrophysics*, 2023, Invited Plenary, 241st AAS Meeting, Seattle.

*Gaussian Processes for EPRV*, 2022, Invited Tutorial, University of Oxford, UK.

*Methods for scalable probabilistic inference*, 2022, Colloquium, University of Illinois Urbana-Champaign.
2022, Colloquium, UC Berkeley.
2022, Colloquium, University of Oxford, UK.
2021, Invited Talk, Institute for Pure & Applied Mathematics, UCLA.

*Advanced probabilistic modeling*, 2021, Tutorial, Harley Wood Winter School of Astronomy, Australia.

*Open-source software for probabilistic data analysis in astronomy*, 2021, Seminar, Instituto de Astrofísica, Portugal.

*Gaussian processes & stellar variability*, 2021, Seminar, CARMENES Team Meeting.

*Extending JAX with custom C++ & CUDA*, 2021, Invited Talk, IRIS-HEP Topical Meeting, CERN.

*Open source software for probabilistic data analysis*, 2020, Invited Talk, OzGrav Early Career Researcher Symposium, Australia.

*The why & how of exoplanet, a domain-specific PyMC3 extension*, 2020, Contributed Talk, PyMC Con.

*A modular ecosystem for probabilistic data analysis*, 2019, Invited Talk, Open Digital Infrastructure in Astronomy conference, Kavli Institute for Theoretical Physics.

*Exoplanet population inference, a tutorial*, 2019, Invited Talk, Exostar19 conference, Kavli Institute for Theoretical Physics.

*Astronomy as a testbed for statistical method development*, 2019, Colloquium, Center for Statistics and Machine Learning, Princeton.

*Data-driven discovery in the astronomical time domain*, 2018, Colloquium, Institute for Theory and Computation, Harvard-Smithsonian Center for Astrophysics.
2018, Colloquium, University of California, Santa Cruz.
2017, Interdisciplinary Colloquium, CIERA, Northwestern University.

*A practical introduction to Gaussian Processes for astronomy*, 2017, Invited Talk, Statistical Challenges in Astrophysics, University of New South Wales, Australia.

*Long-period transiting planets & their population*, 2016, Invited talk, Exoplanets I, Davos.
2016, Invited talk, Statistical Challenges of Modern Astrophysics, Carnegie Mellon.
2016, Colloquium, Villanova.

*Scalable Gaussian processes & the search for transiting exoplanets*, 2015, Data Science at the LHC, CERN, Geneva.

*Discovery & characterization of transiting exoplanets & their population*, 2015, Colloquium, University of Washington.

*Hierarchical inference for exoplanet population inference*, 2015, IAU Symposium, Honolulu.

*Data-driven models*, 2015, Extreme precision radial velocities, Yale.

*Population inference from noisy & incomplete catalogs*, 2015, Local Group Astrostatistics, University of Michigan.

*Time series analysis, Gaussian Processes, and the search for exo-Earths*, 2014, PyData NYC conference, New York.

*Introduction to Gaussian Processes, probabilistic graphical models, and deep learning*, 2014, Astro Hack Week, University of Washington.

*An astronomer's introduction to Gaussian processes*, 2014, Bayesian Computing for

Astronomical Data Analysis (Summer school at Penn State University).

## Grants

NSF-CDS&E (PI: Agol) *Development of fast, multi-dimensional Gaussian Processes for Exoplanet discovery and beyond*, $471,048.00, 2019–2022

NSF-AAG (PI: Agol), *Collaborative Research: Masses and architectures of (potentially habitable) exoplanet systems*, $491,950, 2016–2018

K2 Guest Observer – Cycle 3 (PI: Penny), *Free-Floating and Bound Planet Mass Measurements with K2: Ground- and Space-Based Photometry, Event Detection and Modeling*, $84,000, 2016–2017

K2 Guest Observer – Cycle 3 (PI: Hogg), *Ultra-precise photometry in crowded fields: A self-calibration approach*, $100,000, 2016–2017

XSEDE (PI: Foreman-Mackey), *A systematic search for transiting exoplanets using K2*, 100,000 CPU hours, 2015–2016

## Honors

Kavli Fellow, 2015.

Sagan Postdoctoral Fellowship, 2015–2017.

James Arthur Graduate Fellowship, 2014.

Horizon Fellowship in the Natural & Physical Sciences, 2012.

Henry M. MacCracken Fellowship, 2010.

NSERC Undergraduate Summer Research Award, 2007.

## Professional service & activities

Associate Editor-in-Chief — Journal of Open Source Software, 2019–2025

Active Referee — AAS Journals, MNRAS, PASP, A&A, Journal of Statistical Software, Journal on Uncertainty Quantification, Journal of Open Source Software

Panelist — NSF, NASA, LSSTC